

Build Firewall using FreeBSD

Present by

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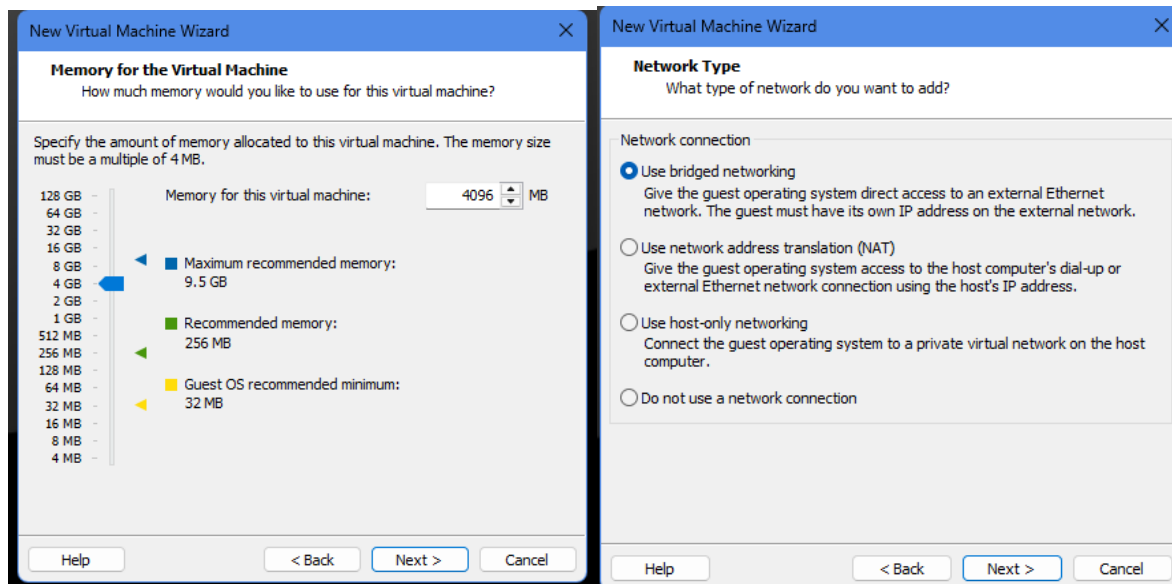
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TJKT XI-3

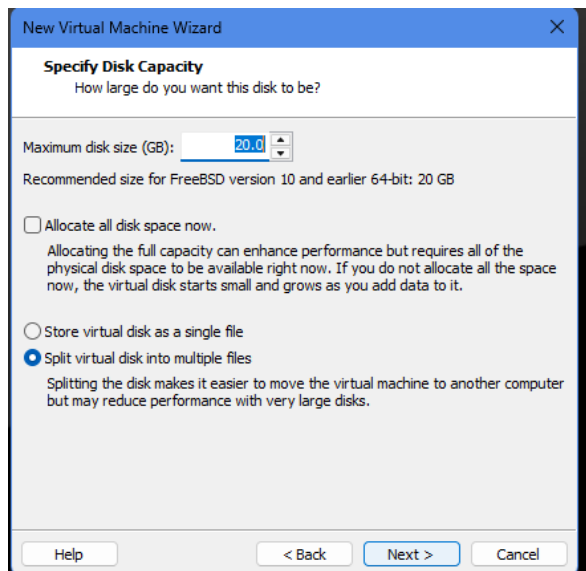
Clcurug 2

1.1 Setup Virtual Machine

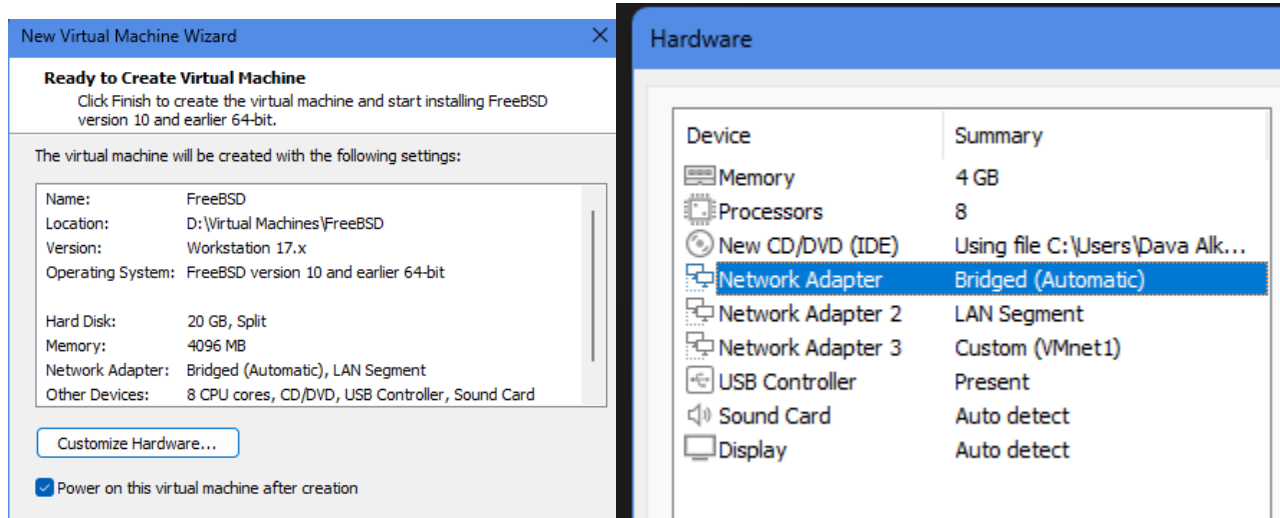
1. Atur ram/memory minimal 2gb (supaya lancer aja) dan gunakan bridge sebagai jalur masuk internet



2. Storage minimal di 10gb



- Choose customize hardware, tambahkan network adapter dan atur na-1 untuk bridge, adapter 2 Lan Segment, *opsional adapter 2 ke host only jika ingin disambungkan ke host*



1.2 Setup FreeBSD Opnsense Installer

- For default login use installer with opnsense password

```
Service 'sysctl' has been restarted.
>>> Invoking start script 'beep'
Root file system: /dev/iso9660/OPNSENSE_INSTALL
Wed Nov 19 09:10:19 UTC 2025

*** OPNsense.internal: OPNsense 25.7 (amd64) ***

LAN (em0)      -> v4: 192.168.1.1/24
WAN (em1)      ->

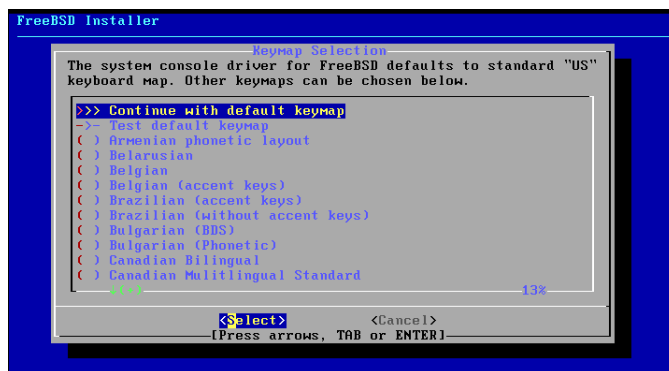
HTTPS: sha256 05 16 DB B8 04 F8 2B 57 FC F8 72 C1 6F 9F 49 51
          C6 7C DB C2 14 74 EE 41 E5 98 01 5E C2 FE 42 D2
SSH:   SHA256 venbFKIMIXGs9JNgHE/SReL3X2uH8ID0EehvNTUGoRs (ECDSA)
SSH:   SHA256 4Kev0s6Ql/f3ScB0onsrKcm51cN4TjMPRIxuj76kxbM (ED25519)
SSH:   SHA256 0Z7tyaTNaU+hb6GJon6R5gFE34hK9kphLT0GuSX41PU (RSA)

Welcome! OPNsense is running in live mode from install media. Please
login as 'root' to continue in live mode, or as 'installer' to start the
installation. Use the default or previously-imported root password for
both accounts. Remote login via SSH is also enabled.

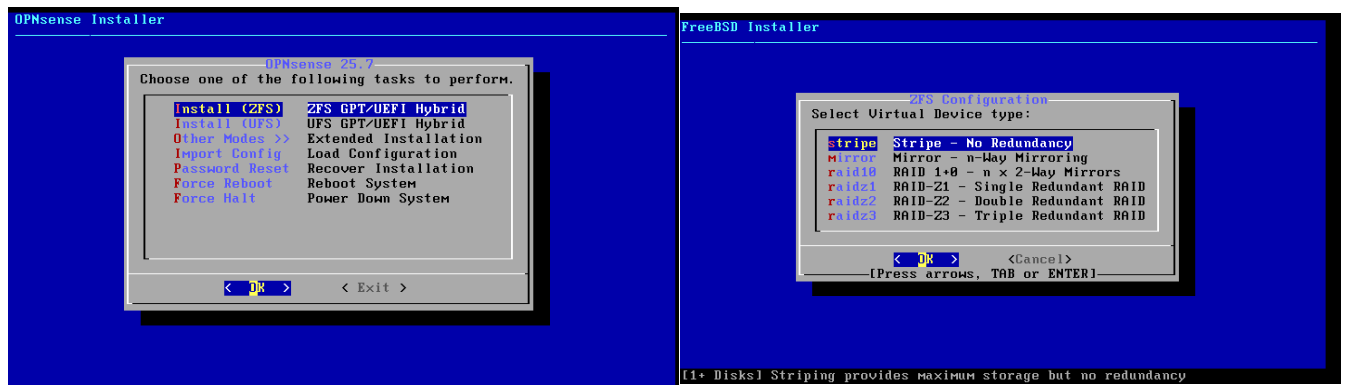
FreeBSD/amd64 (OPNsense.internal) (ttyv0)

login: installer
Password:
```

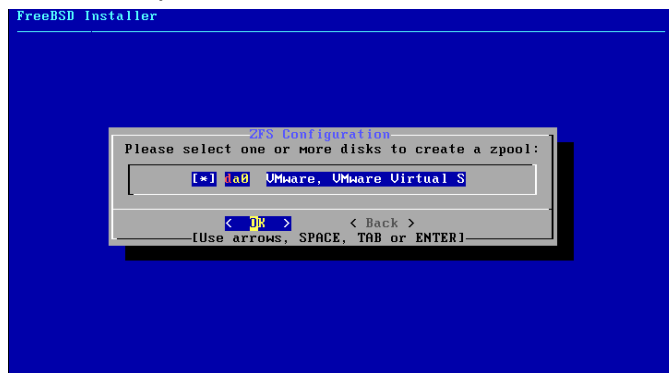
- This option you can direct enter for the keymap



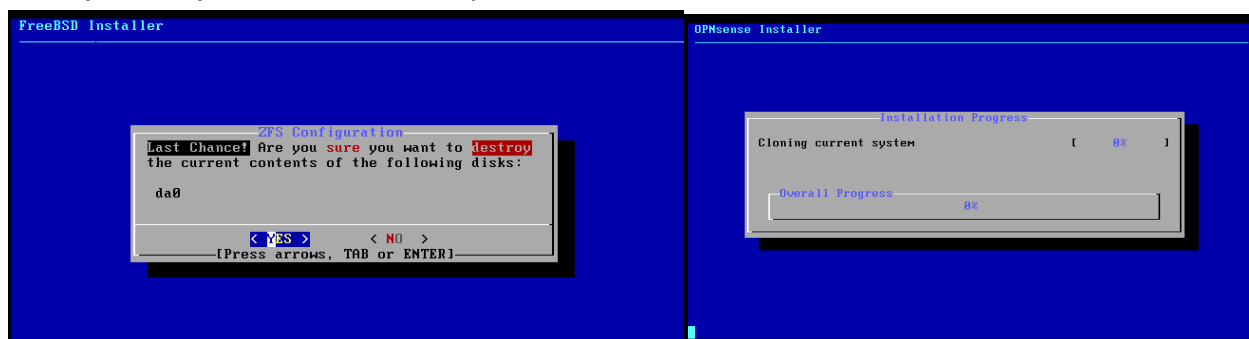
- Select the first option and then enter



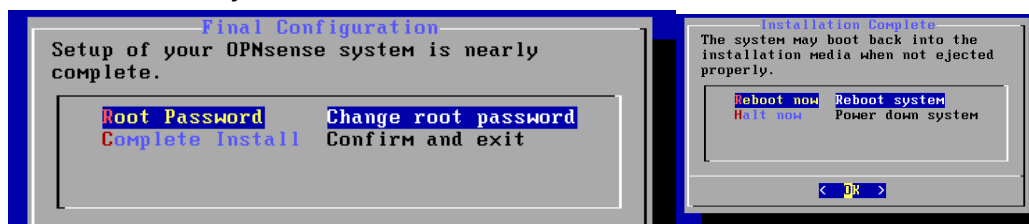
- Press the space for select the virtual disk, after that you can enter for next step



- Click yes and you should wait the process



- You can change the root password or complete install and let the default root password and reboot the system



- If the setup was right you can login with final configuration you set

```
>>> Invoking start script 'newwanip'
>>> Invoking start script 'freebsd'
>>> Invoking start script 'syslog'
>>> Invoking start script 'carp'
>>> Invoking start script 'cron'
Starting Cron: OK
>>> Invoking start script 'openvpn'
>>> Invoking start script 'sysctl'
Service 'sysctl' has been restarted.
>>> Invoking start script 'beep'
Root file system: zroot/ROOT/default
Wed Nov 19 18:21:24 UTC 2025

*** OPNsense.internal: OPNsense 25.7 (amd64) ***

LAN (em0)      -> v4: 192.168.1.1/24
WAN (em1)      ->

HTTPS: sha256 8B 51 FF 0B 1A B2 4D BE 09 83 DA DD 23 54 97 C5
              5D D7 33 21 E5 EE 42 C1 FD 6D E2 E9 C3 DE 4E AB

FreeBSD/amd64 (OPNsense.internal) (ttyv0)

login: root
Password: █
```

1.3 OpenSense Setup

1. We will assign interfaces na2 and 3 and then assign ip address and dhcp server

From the console menu, select opt 1

Skip LAGGs, Skip VLANs

Select em0 for WAN, Select em1 for LAN

```
em2          00:0c:29:d8:da:44 Intel(R) Legacy PRO/1000 MT 82545EM (Copper)

If you do not know the names of your interfaces, you may choose to use
auto-detection. In that case, disconnect all interfaces now before
hitting 'a' to initiate auto detection.

Enter the WAN interface name or 'a' for auto-detection: em0

Enter the LAN interface name or 'a' for auto-detection
NOTE: this enables full Firewalling/NAT mode.
(or nothing if finished): em1

Enter the Optional interface 1 name or 'a' for auto-detection
(or nothing if finished): em2

Enter the Optional interface 2 name or 'a' for auto-detection
(or nothing if finished):

The interfaces will be assigned as follows:

WAN -> em0
LAN -> em1
OPT1 -> em2

Do you want to proceed? [y/N]: █
```

Enter if configuration was right

2. Select opt no 2 in console menu, and makesure WAN interfaces get a ip address

Select no-1 or LAN Interfaces

```
Subnet masks are entered as bit counts (like CIDR notation).
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> ^C
*** OPNsense.internal: OPNsense 25.7 (amd64) ***

LAN (em1)      -> v4: 192.168.1.1/24
OPT1 (em2)     ->
WAN (em0)      -> v4/DHCP4: 172.16.6.171/24

HTTPS: sha256 8B 51 FF 0B 1A B2 4D BE 09 83 DA DD 23 54 97 C5
              5D D7 33 21 E5 EE 42 C1 FD 6D E2 E9 C3 DE 4E AB

0) Logout                7) Ping host
1) Assign interfaces      8) Shell
2) Set interface IP address 9) pfTop
3) Reset the root password 10) Firewall log
4) Reset to factory defaults 11) Reload all services
5) Power off system        12) Update from console
6) Reboot system          13) Restore a backup

Enter an option: 2 █
```

Available interfaces:

```
1 - LAN (em1 - static, track6)
2 - OPT1 (em2)
3 - WAN (em0 - dhcp, dhcp6)
```

Enter the number of the interface to configure: 1 █

3. Configure the ip address and subnet

```
Enter the number of the interface to configure: 1
Configure IPv4 address LAN interface via DHCP? [y/N] n
Enter the new LAN IPv4 address. Press <ENTER> for none:
> 10.3.27.1

Subnet masks are entered as bit counts (like CIDR notation).
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> 24
```

4. Skip IPv6 and enable DHCP server on LAN interfaces

```
For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Configure IPv6 address LAN interface via WAN tracking? [Y/n] n
Configure IPv6 address LAN interface via DHCP6? [y/N] n

Enter the new LAN IPv6 address. Press <ENTER> for none:
>

Do you want to enable the DHCP server on LAN? [y/N] y
```

5. Enter the address range start and end, then you should change the web gui protocol and enable to restore the web gui. After that makesure the LAN interfaces ip address has change

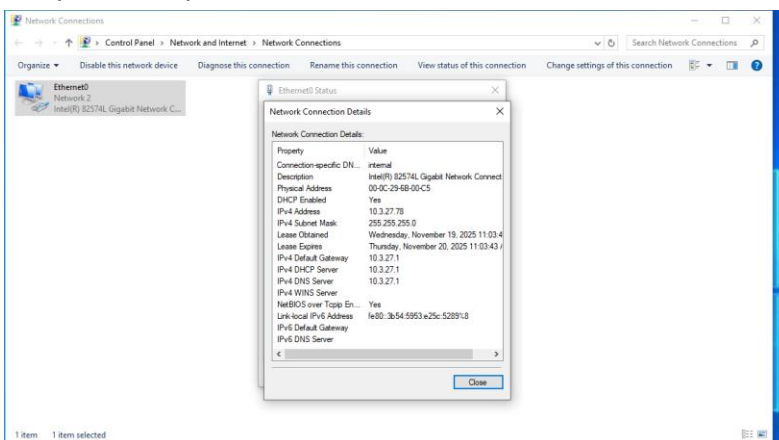
```
Do you want to enable the DHCP server on LAN? [y/N] y

Enter the start address of the IPv4 client address range: 10.3.27.10
Enter the end address of the IPv4 client address range: 10.3.27.200

Do you want to change the web GUI protocol from HTTPS to HTTP? [y/N] y
Restore web GUI access defaults? [y/N] y
```

```
LAN (em1)      -> v4: 10.3.27.1/24
OPT1 (em2)     ->
WAN (em0)      -> v4/DHCP4: 172.16.6.171/24
```

6. On your windows virtual machine, check the ip address using WIN+R and type ncpa.cpl, check the adapter and ip address was reachable

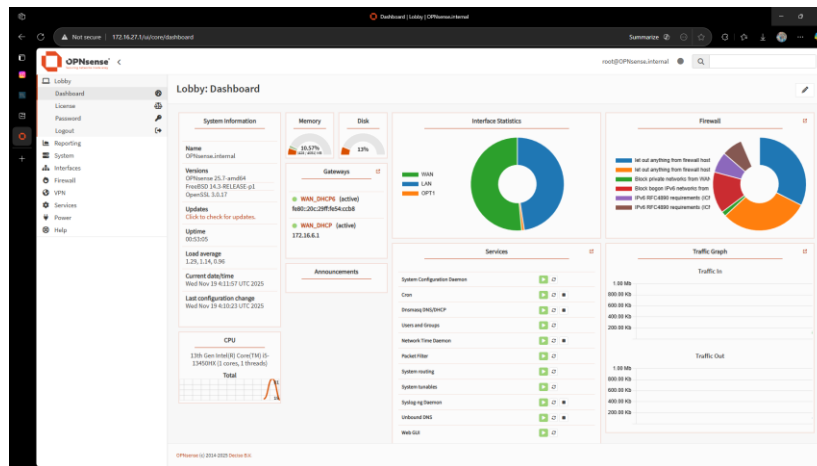


1.4 Opensense and Zenarmor web gui set up

in purpose I should to block the mikrotik.com, Tokopedia and facebook

1. Login to chrome/edge/whatever browser you have

Search the ip address LAN interfaces



2. You should update the version, you can see your version currently and you can also see the version was release in dashboard, if you wanna update using shell or console, freebsd avail the option, you can use the opt no.12 or using bsd and type pkg upgrade/update -f

Lobby: Dashboard

System Information

Name
OPNsense.internal

Versions
OPNsense 25.7-amd64
FreeBSD 14.3-RELEASE-p1
OpenSSL 3.0.17

Updates
[Click to check for updates.](#)

Uptime
00:03:33

Load average
1.10, 0.64, 0.28

Current date/time
Wed Nov 19 4:22:17 UTC 2025

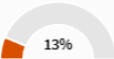
Last configuration change
Wed Nov 19 4:10:23 UTC 2025

Memory



12.07%
(489 / 4051) MB

Disk



13%

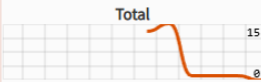
Gateways

- WAN_DHCP6** (active)
fe80::20c:29ff:fe4f:33c5
- WAN_DHCP** (active)
172.16.6.1

CPU

13th Gen Intel(R) Core(TM) i5-13450HX (8 cores, 8 threads)

Total



15
8

Re: OPNsense 25.7.7 released
Thu, 06 Nov 2025 20:22:11 GMT
A hotfix release was issued as 25.7.7_2:

o ui: bootgrid: prevent full table redraw without onDataProcessed trigger
o ui: bootgrid: add missing datakeys to two pages

o system: fix a HA sync regression introduced in 25.7.6 that prevented a sync from succeeding in an edge case

```

py311-sniffio-1.3.1
py311-socksio-1.0.0_1
py311-sortedcontainers-2.4.0_1
py311-tzdata-2025.2
py311-vici-5.9.11_1
radvd-2.20
readline-8.2.13_2
rrdtool-1.9.0_1 (direct dependency changed: perl5)
samplerator-1.3.8.r1_1
zip-3.0_4
zstd-1.5.7

Number of packages to be upgraded: 78
Number of packages to be reinstalled: 94

The process will require 8 MiB more space.
197 MiB to be downloaded.

Proceed with this action? [y/N]: y
Please type 'Yes!' or 'No!' to make a selection

Proceed with this action? [y/N]: y
[1/171] Fetching py311-sqlite3-3.11.14_1.pkg: 100%   40 KiB   40.0kB/s      00:01
[2/171] Fetching py311-anyio-4.11.0.pkg: 16%    39 KiB    0.0kB/s      00:57 ETA
```

```
Status      Settings  Changing  Updates  Plugins  Packages

wpa_supplicant: 2.11.5 -> 2.11.7

Installed packages to be REINSTALLED:
  cyrus-sasl-2.1.28.5 (provided shared library changed)
  cyrus-sasl-ssapi-2.1.28 (provided shared library changed)
  dnsmasq-2.91.1 (required shared library changed)
  glib-2.84.1_3,2 (required shared library changed)
  ntp-4.2.8p18.4 (direct dependency changed: perl5)
  openssl26-client-2.6.10 (required shared library changed)
  rtdtool-1.9.0.1 (direct dependency changed: perl5)

Number of packages to be upgraded: 78
Number of packages to be reinstalled: 7

The process will require 8 MiB more space.
140 MiB to be downloaded.
[1/76] Fetching openssl-3.0.18.1.pkg: ..... done
[2/76] Fetching php83-curl-8.3.27.pkg: ..... done
[3/76] Fetching boost-libs-1.89.0.1.pkg: ..... done
[4/76] Fetching py311-numpy-1.26.4_7.1.pkg: ..
```

System: Configuration: Wizard

[Welcome](#) | [General Information](#) | [Network \[WAN\]](#) | [Network \[LAN\]](#)

1

Hostname

Dava

2

Domain

internal

3

Language

English

4

Timezone

Asia/Jakarta

5

DNS Servers

8.8.8.8 × 1.1.1.1 ×
[Clear All](#) [Copy](#) [Text](#)

6

Override DNS

☒

7

DNS [Unbound]

1

Enable Resolver

☒

2

Enable DNSSEC Support

☐

3

Harden DNSSEC data

☐

4. In system firmware, go to plugins and install sunnyvalley/sensei

os-sunnyvalley	1.5_1	2.44KiB	2	OPNsense	Vendor Repository for Zenarmor (Enterprise Security Modules - NGFW, SSE, SASE, f.k.a Sensei)	1	+	
----------------	-------	---------	---	----------	--	---	---	--

5. After you install sunnyvaley, search sensei and download all

Status	Settings	Changelog	Updates	Plugins	Packages			
Name	Version	Size	Tier	Repository	Comment	☐ Show community plugins		
os-sensei (installed)	2.2	266MiB	2	SunnyValley	Enterprise Security Extensions for OPNsense (ZENARMOR)		1	🗑
os-sensei-agent (installed)	2.2	118MiB	2	SunnyValley	ZENARMOR Connectivity Agent for Cloud Central Management		1	🗑
os-sensei-updater (installed)	1.18	4.09KiB	2	SunnyValley	OPNsense ZENARMOR Plugin Updater		1	🗑
os-sunnyvalley (installed)	1.5_1	2.44KiB	2	OPNsense	Vendor Repository for Zenarmor (Enterprise Security Modules - NGFW, SSE, SASE, f.k.a Sensei)		1	🗑

6. Look at your side panel and open zenarmor, click the column, I agree and next

Lobby

Reporting

System

Interfaces

Firewall

VPN

Services

Zenarmor

Dashboard

Reports

Live Sessions

Policies

Settings

Notifications

Power

Help

1 Welcome

2 Database Settings

3 Interface Settings

4 Subscription

5 Finish

Zenarmor

You're ready to do the initial configuration.

☐ Check here to indicate that you have read and agree to the [Terms of Service](#) and [Privacy Policy](#).

Uninstall

I Agree

Want more power?

Zenconsole is the cloud central manager for Zenarmor.

[Connect to Zenconsole](#)

Have feedback?

We'd love to hear your feedback on Zenarmor.

[Send Feedback](#)

7. Hardware check and select database, select sqlite and choose support 100 devices

1 Welcome

2 Database Settings

3 Interface Settings

4 Subscription

5 Finish

☐ Install a local MongoDB Database

☐ Use a Remote Elasticsearch Database

☐ Use a local SQLite Database (Support up to 100 devices)

Low-end hardware

We have detected a low-end hardware. No worries, zenarmor will offer an optimum configuration for you.

CPU Model: Common KVM processor

CPU Score: 505181

Physical Memory Size: 2.4 GB

Previous

Next

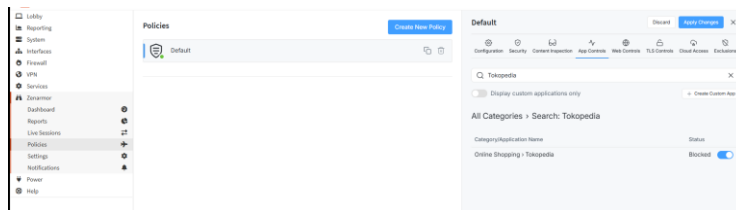
8. Better you follow me

after that, click get me free edition, and you

don't have to fill in the email

1.5 setup a policies using Zenarmor and Firewall rules

1. Look at your side panel, click Zenarmor, policies, default and search Tokopedia



Do the same thing as you did before (Facebook)

All Categories > Search: Facebook	
Category/Application Name	Status
Social Network > Facebook	Blocked ☒
Social Network > Facebook Apps	Blocked ☒
VOIP > Facebook Call	Blocked ☒
Cloud Services > Facebook CDN	Blocked ☒
Instant Messaging > Facebook Chat	Blocked ☒
Social Network > Facebook Events	Blocked ☒
Social Network > Facebook Friends	Blocked ☒
Gaming > Facebook Games	Blocked ☒
Social Network > Facebook Groups	Blocked ☒
Social Network > Facebook Like	Blocked ☒
Social Network > Facebook Message	Blocked ☒
Instant Messaging > Facebook Messenger	Blocked ☒
Social Network > Facebook Mobile	Blocked ☒
Social Network > Facebook Notes	Blocked ☒
Social Network > Facebook Notification	Blocked ☒
Social Network > Facebook Photos	Blocked ☒
Social Network > Facebook Photos and Videos	Blocked ☒

2. Look at your side panel, click firewall, click aliases, add a new

Edit Alias

 Enabled



 Name

 Type

Host(s)

 Categories

 Content

 Clear All  Copy  Text

 Statistics



 Description